Department of Electrical Engineering, Faculty of Engineering
Rajamangala University of Technology Thanyaburi
Pathum Thani 12110, Thailand

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Editor-in-Chief
IEEE Access

Subject: Submission of the manuscript "Two-Time-Scale Coordination of Vehicle-to-Grid Voltage Support and Frequency Reserve in a Two-Area System Using Proximal Policy Optimization"

Dear Editor,

We are pleased to submit the above manuscript, an original research article, for consideration in *IEEE Access*.

What the paper does. An aggregated fleet of vehicle-to-grid electric vehicles can support distribution voltages and provide frequency reserve, but both services draw on the same converter rating and stored energy. The paper formulates their coordination as a two-time-scale control problem: every hour each area aggregator chooses a fleet power set-point and the share of its plugged-in converter rating held as frequency reserve. The set-points are evaluated by AC power flow on IEEE 33-bus feeders in two areas with an energy-consistent fleet model; the reserve sets the gain and limit of a V2G droop loop in a dynamic two-area load-frequency control model with inertia, governors, a tie line, and automatic generation control, simulated at a 20-ms step. A proximal policy optimization agent is trained with 5 independent seeds and compared, under identical conditions, with uncontrolled charging, a rule-based volt-watt and droop controller, and a day-ahead optimization with perfect foresight.

Main findings. On 30 held-out days the learned policy lowers the daily performance cost by 11.0%, 44.8%, and 71.6% relative to the three benchmarks, reduces undervoltage by 13.5% and the mean frequency nadir by 5.0% relative to the day-ahead optimization, and keeps its advantage when that benchmark is re-optimized for every reserve share and even against a per-day oracle choice of the share. The paper also reports what the method does not achieve: some hours of the weak residential feeder stay below 0.95 p.u. with every controller, a downward activation of the committed reserve can push further hours below the band, and consecutive-day operation shows a downward drift of the residential fleet energy that a soft terminal penalty does not prevent. These limits are quantified in the manuscript rather than left to the reader.

Why the work suits *IEEE Access*. The paper combines power-system modeling (verified AC power flow and a dynamic LFC model in physical units), reinforcement learning with a stated training and evaluation protocol, and an unusually complete reproducibility package. The energy exchanged by the droop loop is included in the fleet energy balance and checked against exact 3600-s simulations; every number in the manuscript is recomputed from the raw result files by an independent verification script (84 checks); and the code, trained policies, result files, checksums, and exact package versions are released at `https://github.com/kenoto99/v2g-ppo-two-area` (archived on Zenodo, DOI 10.5281/zenodo.23851585).

Declarations. The manuscript is original, has not been published in whole or in part, and is not under consideration by any other journal or conference. All authors have read and approved the submitted version, agree to its submission to *IEEE Access*, and meet the IEEE authorship criteria. The authors declare no conflict of interest. No human or animal subjects were involved. The work was supported in part by Rajamangala University of Technology Thanyaburi.

We would be grateful if the manuscript could be handled by an Associate Editor in the area of power and energy systems or smart grids. Thank you for considering our submission.

Sincerely,

Krischonme Bhumkittipich (Senior Member, IEEE)
Corresponding author, on behalf of all authors
Professor, Department of Electrical Engineering, Faculty of Engineering
Rajamangala University of Technology Thanyaburi, Pathum Thani 12110, Thailand
E-mail: krischonme.b@en.rmutt.ac.th

Co-authors: Sontaya Manmai, Prakasit Prabpal, Yuttana Kongjeen, Kulsomsup Yenchamchalit, Somchat Sonasang, and Kritphon Phanrattanachai.